

# Jimmy Van Hout

[vanhout@umd.edu](mailto:vanhout@umd.edu)

[jimmyvanhout.com](http://jimmyvanhout.com) | [github.com/JimmyVanHout](https://github.com/JimmyVanHout) | [linkedin.com/in/JimmyVanHout](https://linkedin.com/in/JimmyVanHout)

Arlington, VA

## Education

University of Maryland, College Park, MD, *May 2022*

- **Bachelor of Science in Computer Science**
- Minor in Spanish Language, Culture, and Professional Contexts

## Experience

### Embedded Software Engineer

Startup, *March 2026 - Present*

- Work with an experienced electrical engineer to design software and hardware of a medical device for use in hospitals and nursing homes
- Develop software to control mechanical devices critical for patient health and safety
- Assist in selecting and designing hardware, including PCB, to enable software control and proper functioning of device
- Develop software on embedded computer or remote server to store, visualize, and process data

### Embedded Software Engineer

EmbedTek (Subcontractor), *February 2025 - March 2026*

- Worked directly with President & CTO of a medical device manufacturer to design software and hardware for use in hospitals and nursing homes
- Processed data from multiple sensors using multithreading on both cores of an embedded ESP32 microcontroller to achieve high throughput
- Assisted in selecting and designing hardware, including PCB, to enable software control and proper functioning of device

### Full-Stack Software Engineer

Quadrofoil (Contractor), *August 2025 - September 2025*

- Worked with client to identify and solve an issue, gather requirements, obtain product feedback from stakeholders, and deliver a product according to clients' requirements
- Designed website according to client's specifications

## Full-Stack Software Engineer

ScheduleAI, *June 2023 - January 2025*

- Developed a SaaS web application and artificial intelligence algorithm for automated employee scheduling
- Developed backend using Python, Flask, PostgreSQL, Amazon Cognito API; frontend using JavaScript, HTML, CSS
- Implemented server-side user authentication and authorization, including the use of OAuth 2.0, Amazon Cognito, JWTs, and cryptographic libraries
- Develop HTTP REST APIs for programmatic access of data

## Software Engineer

Orangetheory Fitness (Consultant), *August 2022 - November 2022*

- Developed software to analyze and visualize customer data and resource utilization to optimize resource allocation and reduce expenses

## Technical Skills

- **Programming Languages:** Python, JavaScript, C, C++, Java, C#
- **Web:** Python, Flask, JavaScript, Node.JS, HTML, CSS, React, Vue, Gunicorn, NGINX
- **Embedded:** ESP32, STM32, FreeRTOS, esp-idf, Arduino, Raspberry Pi, Linux
- **Cloud Computing/SaaS:** AWS, GCP, Cloudflare, Azure, Stripe
- **Security:** Authentication and authorization, OAuth 2.0, JWTs, cryptographic libraries
- **Parallel and Multithreaded Programming:** HPC, MPI, OpenMP, CUDA
- **Databases:** PostgreSQL, SQLite

## Selected Technical Projects

### Temperature and Pressure Sensor Program

*C, Python*

Read temperature and pressure from a MS5839-02BA sensor via I2C with an ESP32 micro-controller and output the data over serial.

GitHub: [https://github.com/JimmyVanHout/temperature\\_pressure\\_sensor](https://github.com/JimmyVanHout/temperature_pressure_sensor)

### Youth Health Data Tool and Website | [youthhealthstatistics.com](https://youthhealthstatistics.com)

*Python, JavaScript, HTML, CSS*

A tool to retrieve data and generate statistics on youth health and an accompanying website to visualize the data.

GitHub: [https://github.com/JimmyVanHout/youth\\_health\\_data\\_tool](https://github.com/JimmyVanHout/youth_health_data_tool)